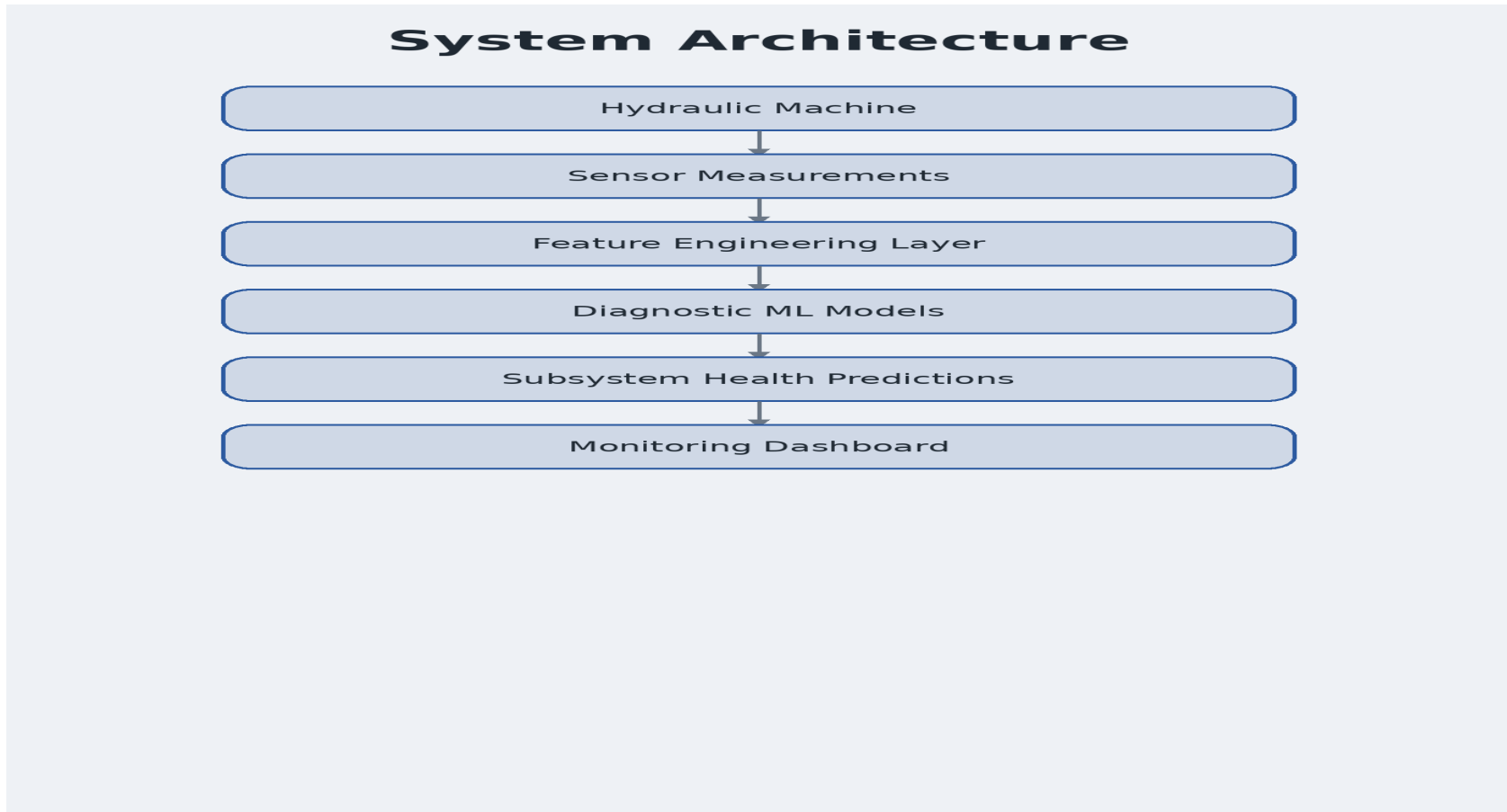


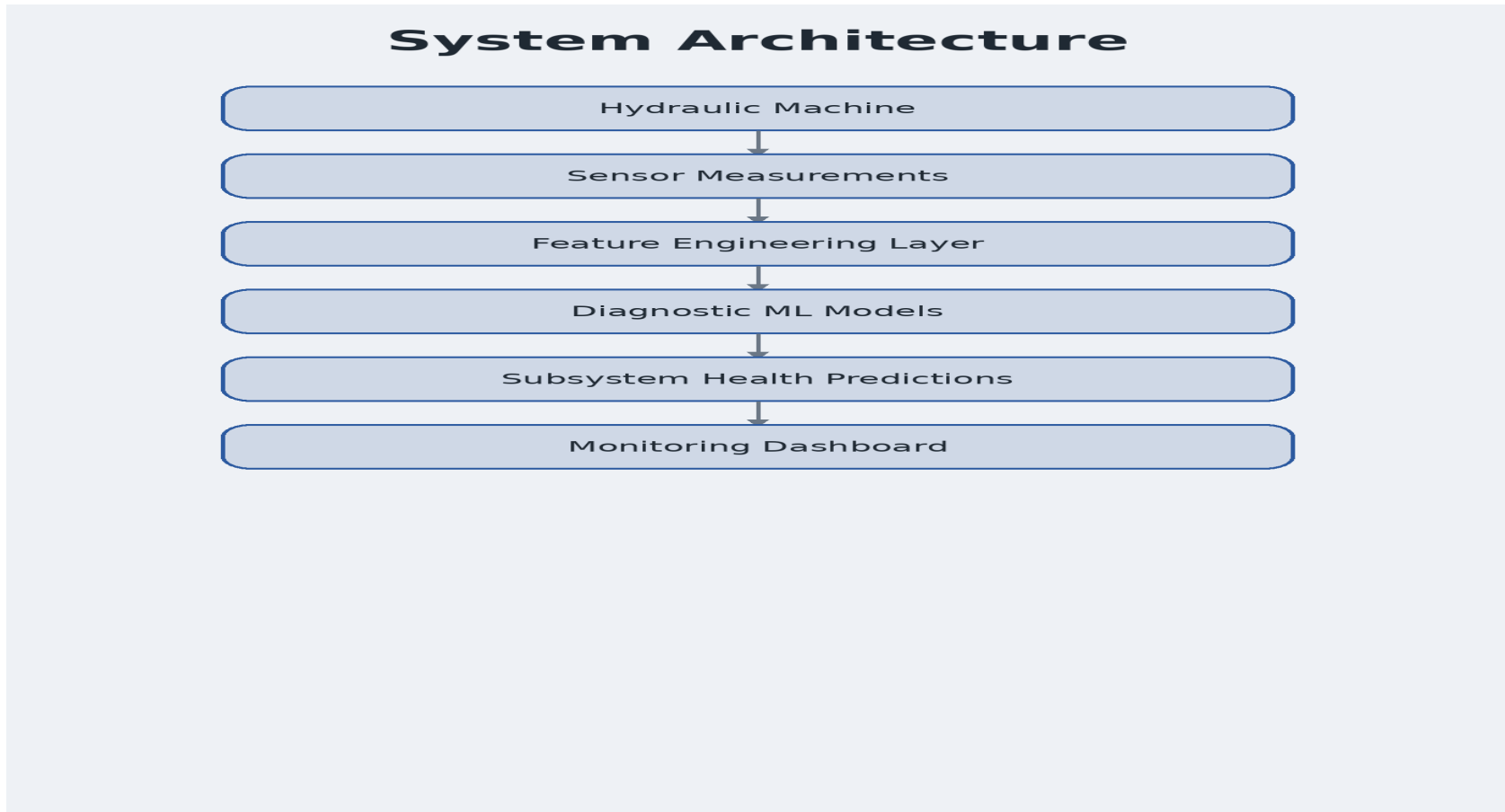
1. Hydraulic Condition Monitoring Using Machine Learning

Predictive maintenance system for hydraulic machinery using subsystem-specific machine learning models and 130k+ engineered sensor features.



2. Why This Problem Matters

Unexpected hydraulic failures increase downtime, cost, and safety risk. The project predicts failures at the subsystem level before breakdown occurs.



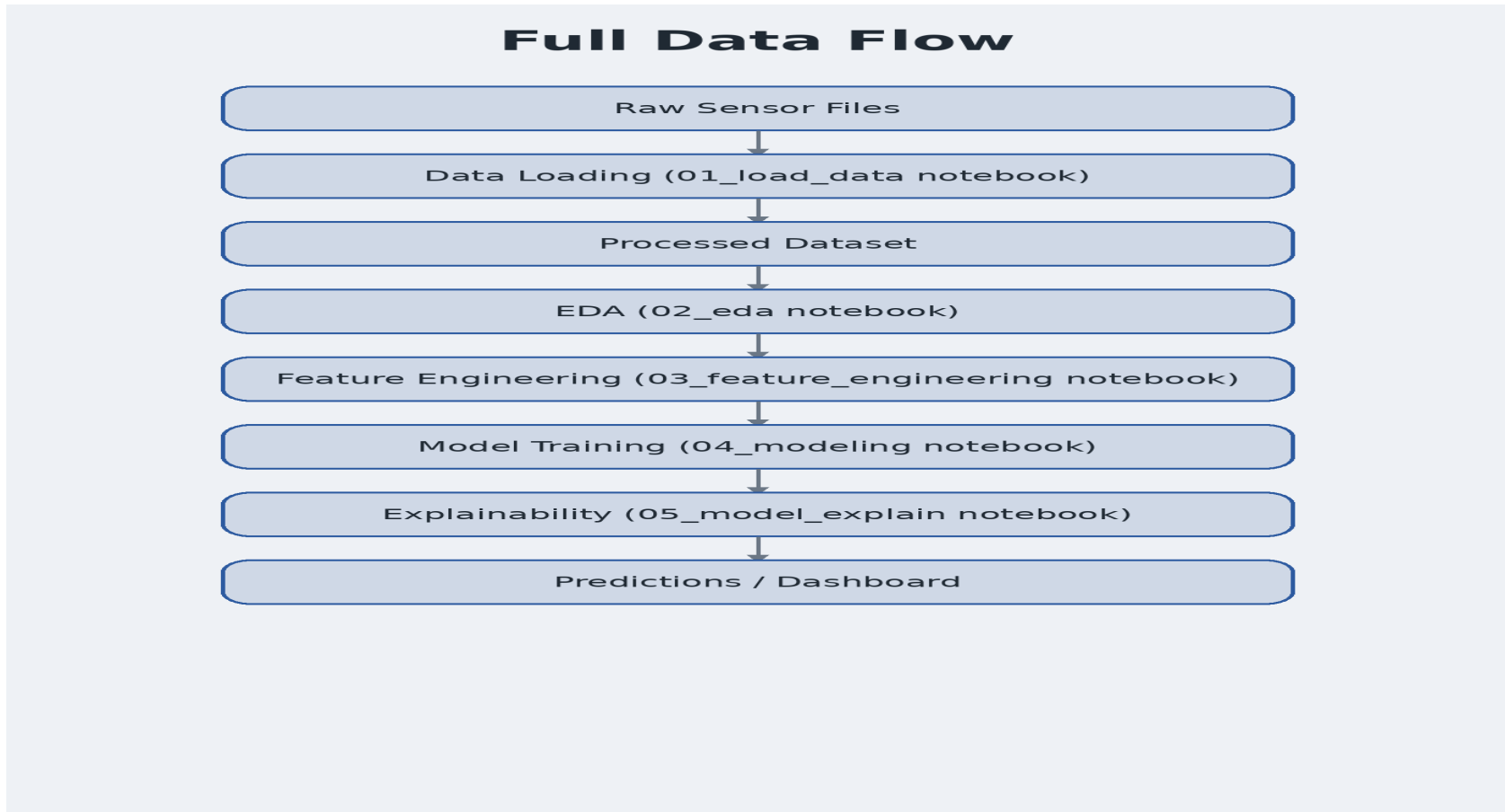
3. Dataset and Sensor Structure

The dataset combines pressure, temperature, vibration, flow, and power measurements across 2,205 operating cycles. Five targets are predicted independently.

Target	Type	Purpose
Cooler Condition	Multiclass	Cooling system health
Valve Condition	Multiclass	Valve degradation level
Pump Leakage	Multiclass	Pump leakage severity
Accumulator Pressure	Multiclass	Accumulator pressure state
Stable Flag	Binary	Overall system stability

4. End-to-End Project Workflow

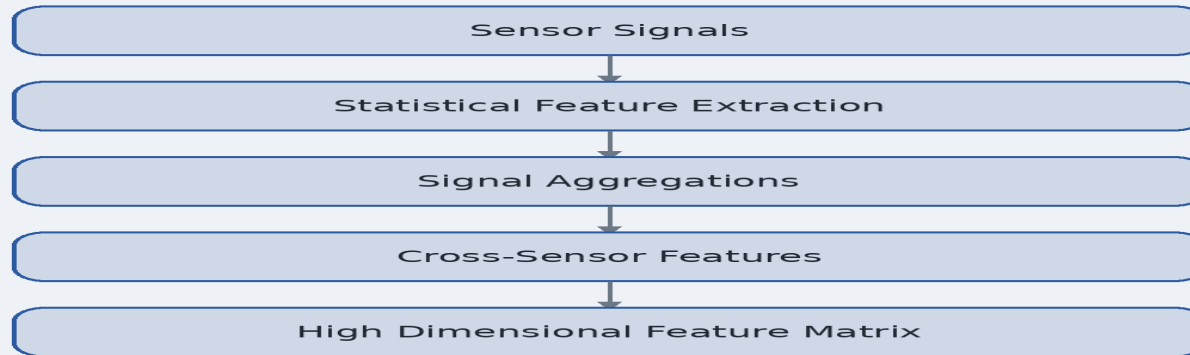
The process is fully reproducible and follows the same notebook order documented in the repository.



5. Feature Engineering Pipeline

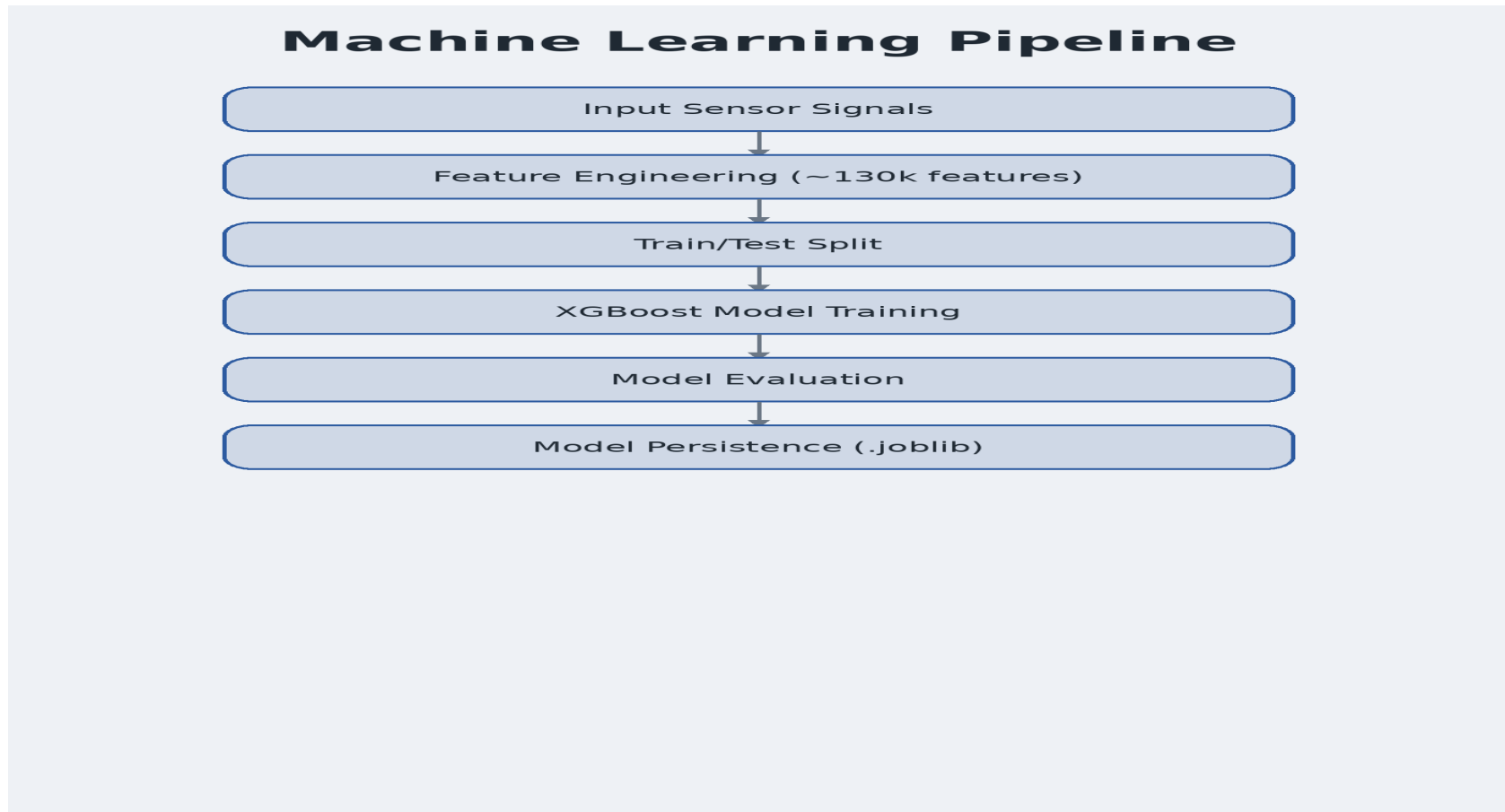
Raw sensor signals are transformed into more than 130,000 engineered features. Approximately 500 of the most informative features are retained before model training.

Feature Engineering Pipeline



6. Machine Learning Pipeline

The final modeling pipeline uses train/test split, XGBoost training, evaluation, and joblib persistence for deployment.



7. Subsystem Model Architecture

Rather than using a single global model, one model is trained per subsystem. This improves interpretability, retraining, and maintenance value.

Subsystem Model Architecture



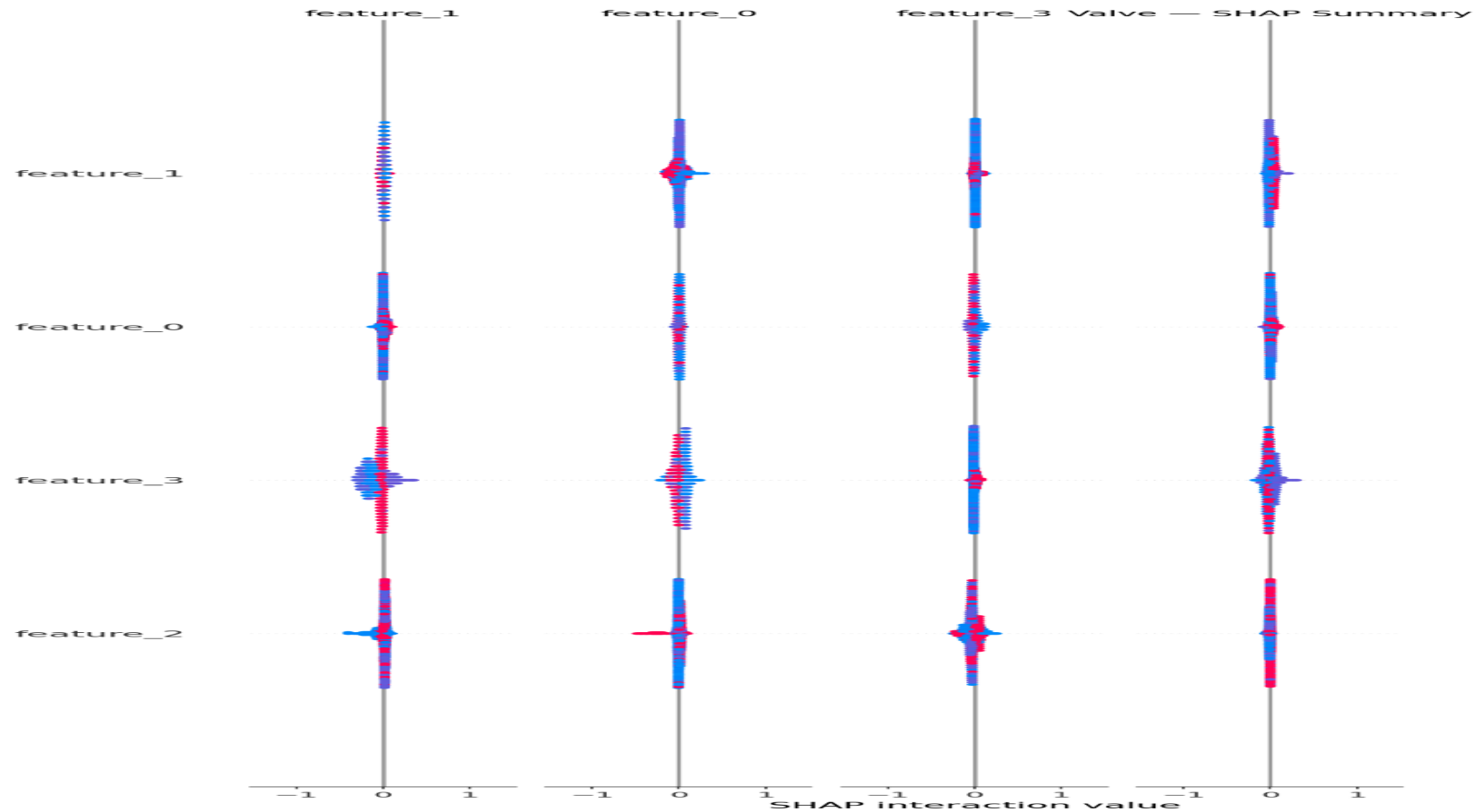
8. Model Performance

The strongest models are Cooler and Pump. Valve remains the most difficult target because its signal is less separable.

Subsystem	Accuracy	Interpretation
Cooler	0.995	Strong temperature signature
Valve	0.659	Weak and overlapping signal
Pump	0.956	Clear leakage pattern
Accumulator	0.943	Reliable pressure features
Stable	0.943	Strong global separation

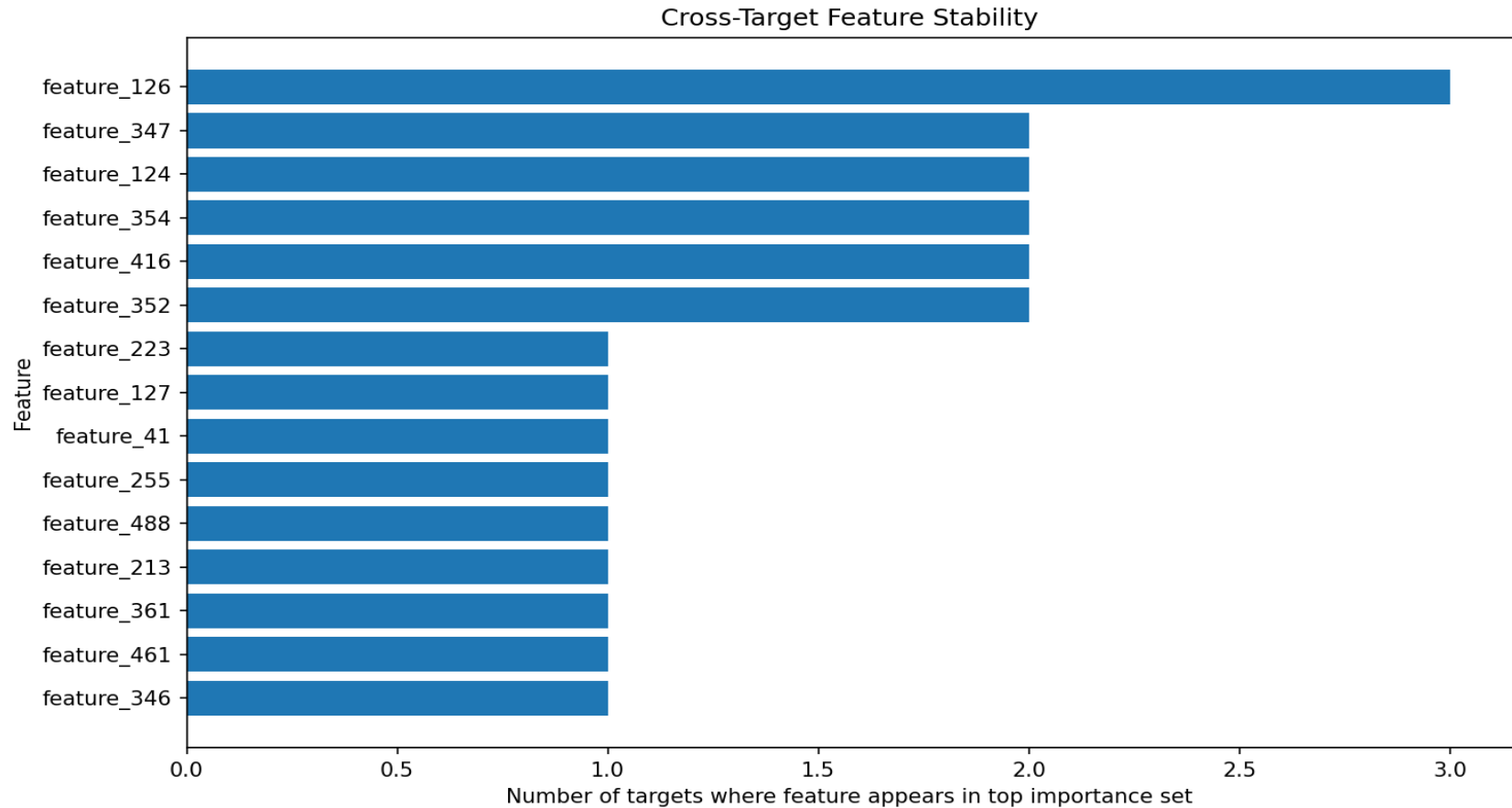
9. Critical Insight: Why Valve Is Harder

Valve degradation produces a weaker physical signal than the other targets. SHAP confirms that the model is learning useful patterns, but the classes overlap significantly.



10. Explainable Machine Learning

SHAP and cross-target feature stability analysis show that pressure and vibration features consistently drive predictions across subsystem models.



11. Deployed Monitoring Dashboard

The Streamlit application loads feature-engineered CSV files, validates schema, generates predictions, and exports downloadable results.

Deploy

Hydraulic Condition Monitoring Dashboard

Upload a feature-engineered CSV to generate subsystem condition predictions.

This app predicts:

- Cooler condition
- Valve condition
- Pump leakage
- Accumulator pressure condition
- Stable / unstable state

Input Source

Choose input method

☒ Upload CSV ☐ Use demo sample

Upload your feature-engineered CSV

 Drag and drop file here
Limit 200MB per file • CSV

Browse files

Choose a demo sample or upload a CSV file to begin.

12. Prediction Flow and Business Impact

The final deployed system supports real operational use by producing subsystem predictions, confidence scores, and downloadable CSV outputs for maintenance teams.

Deploy

EPS1_t476
EPS1_t477
EPS1_t478
EPS1_t479
EPS1_t480

Input schema validation passed.

Predictions

	Cooler_Condition_pred	Cooler_Condition_label	Cooler_Condition_confidence	Valve_Condition_pred	Valve_Condition_label	Valve_Condition_confidence	Pump_Leakage_pred	Pump_Leakage_label	Pump_L
0	0	0	0.9989	3	3	0.9987	0	0	

Download Results

Download predictions only (CSV)

Download compact output (CSV)